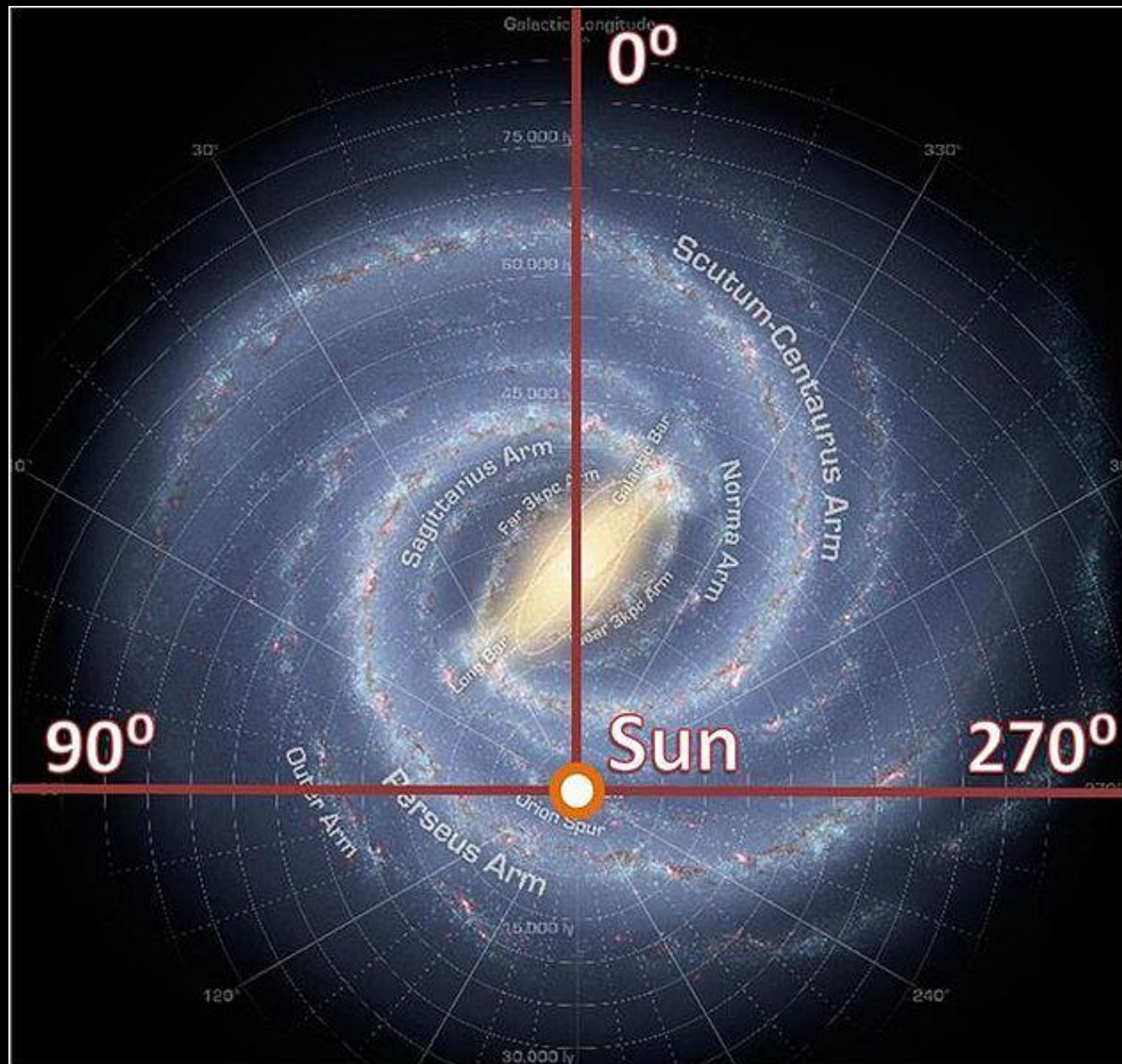
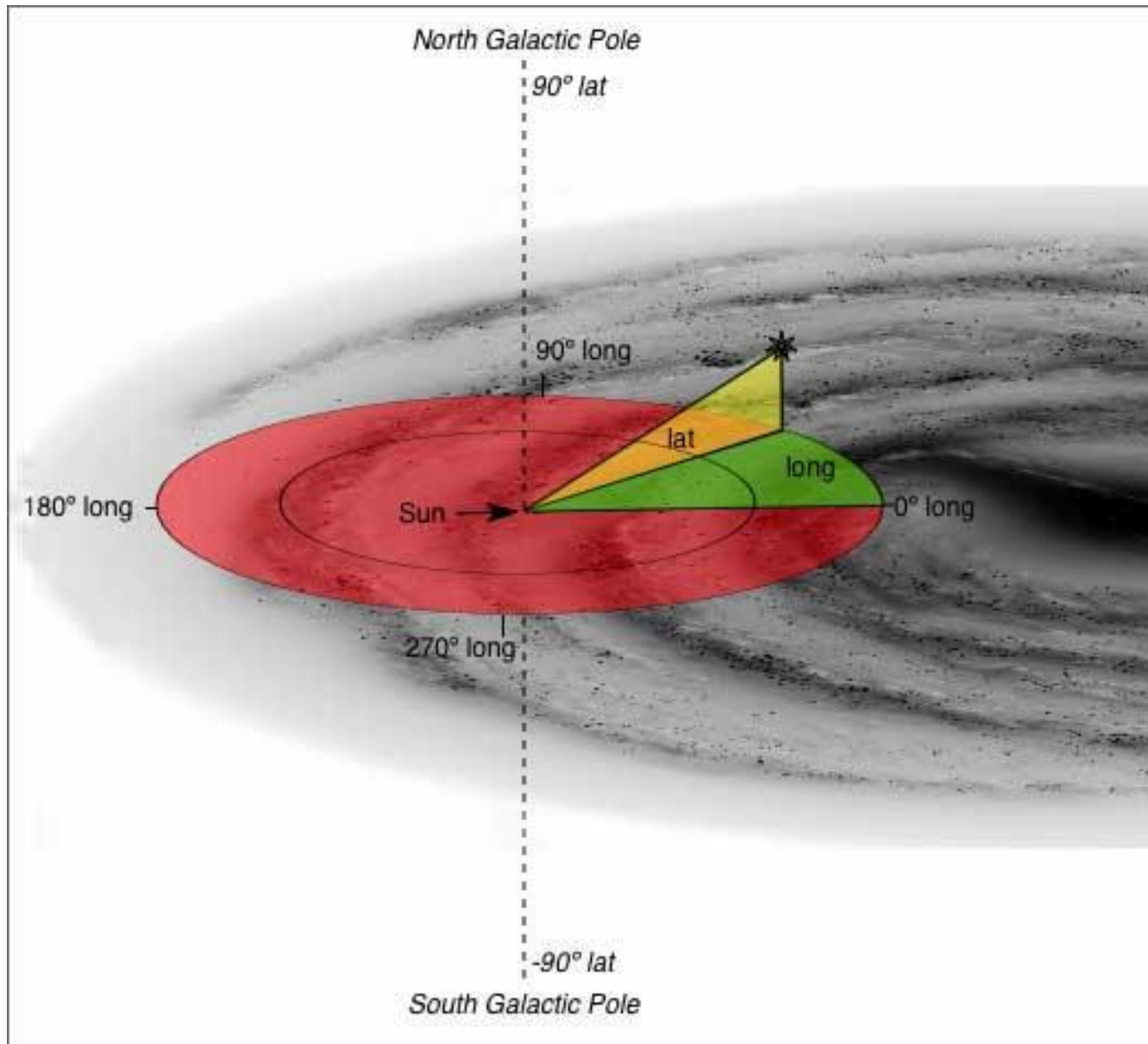
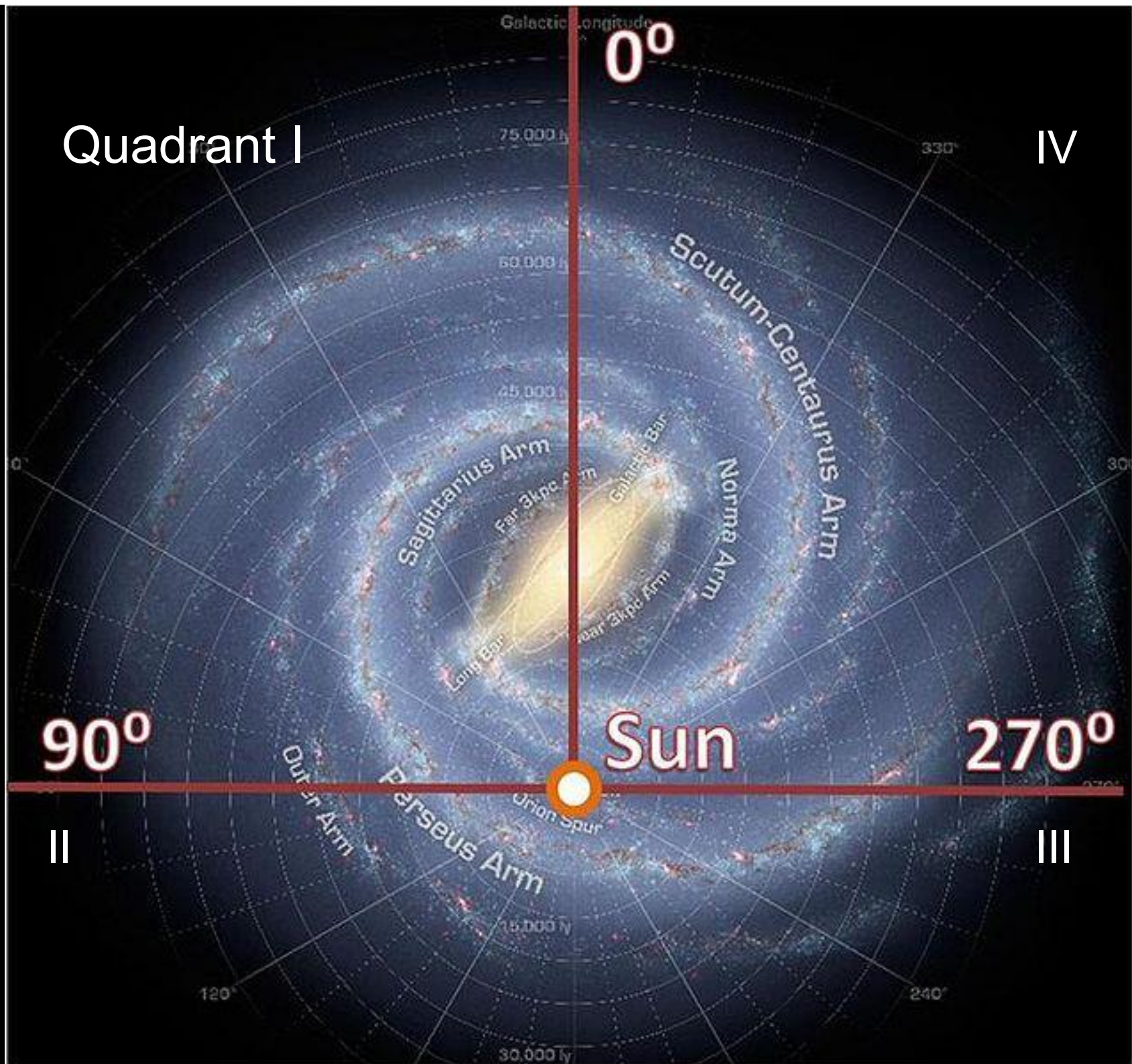


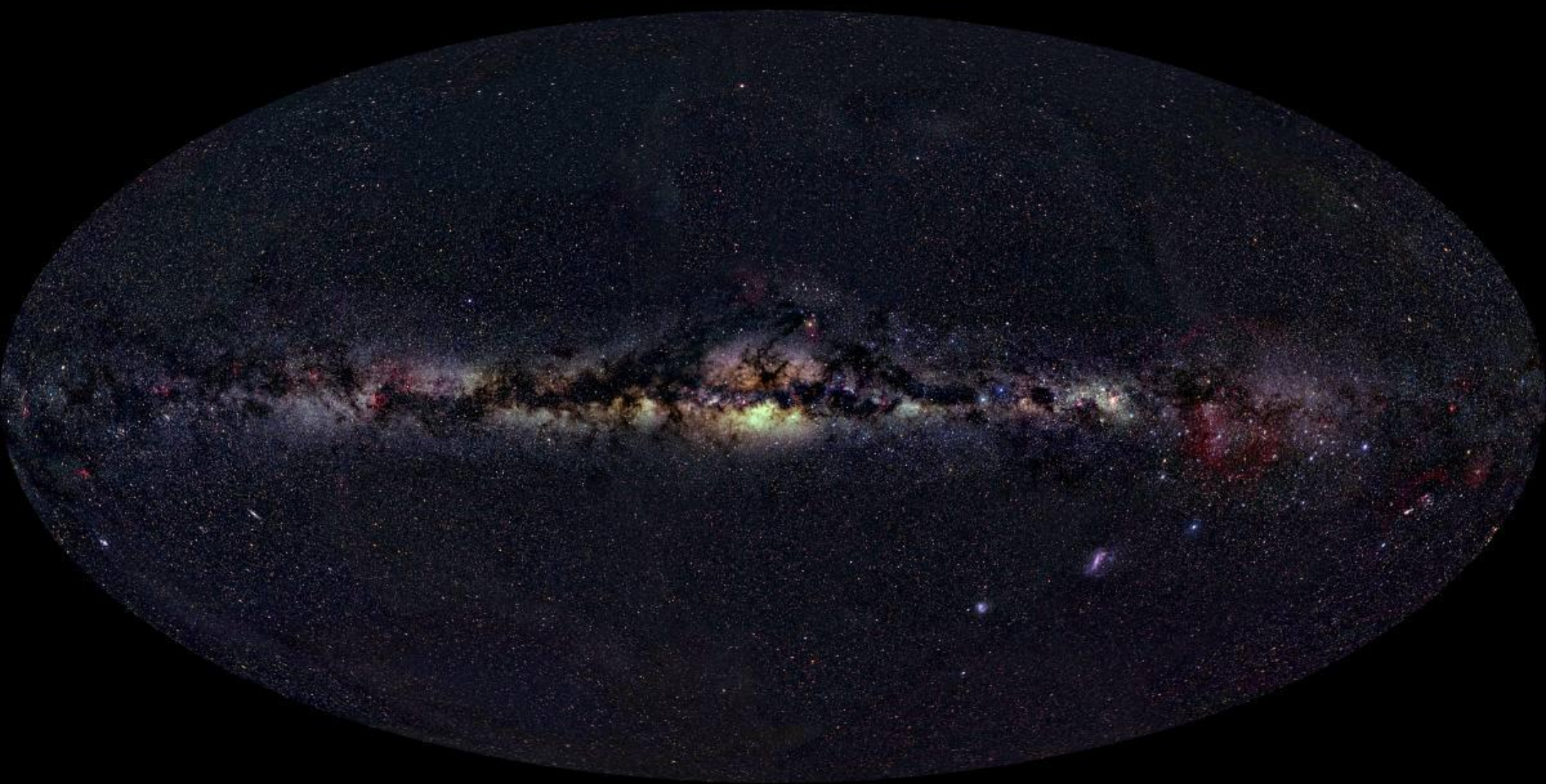
Galactic Rotation

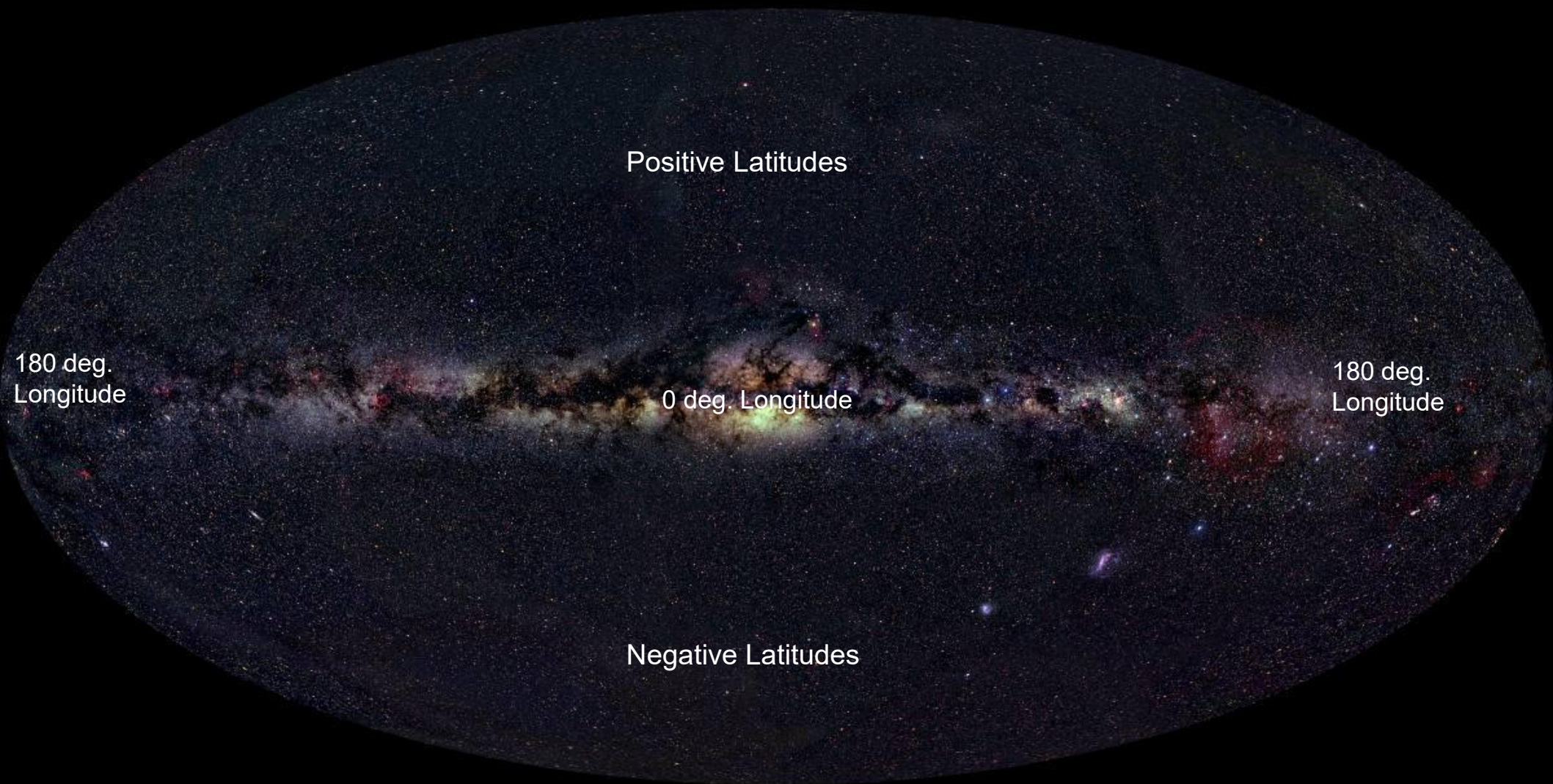


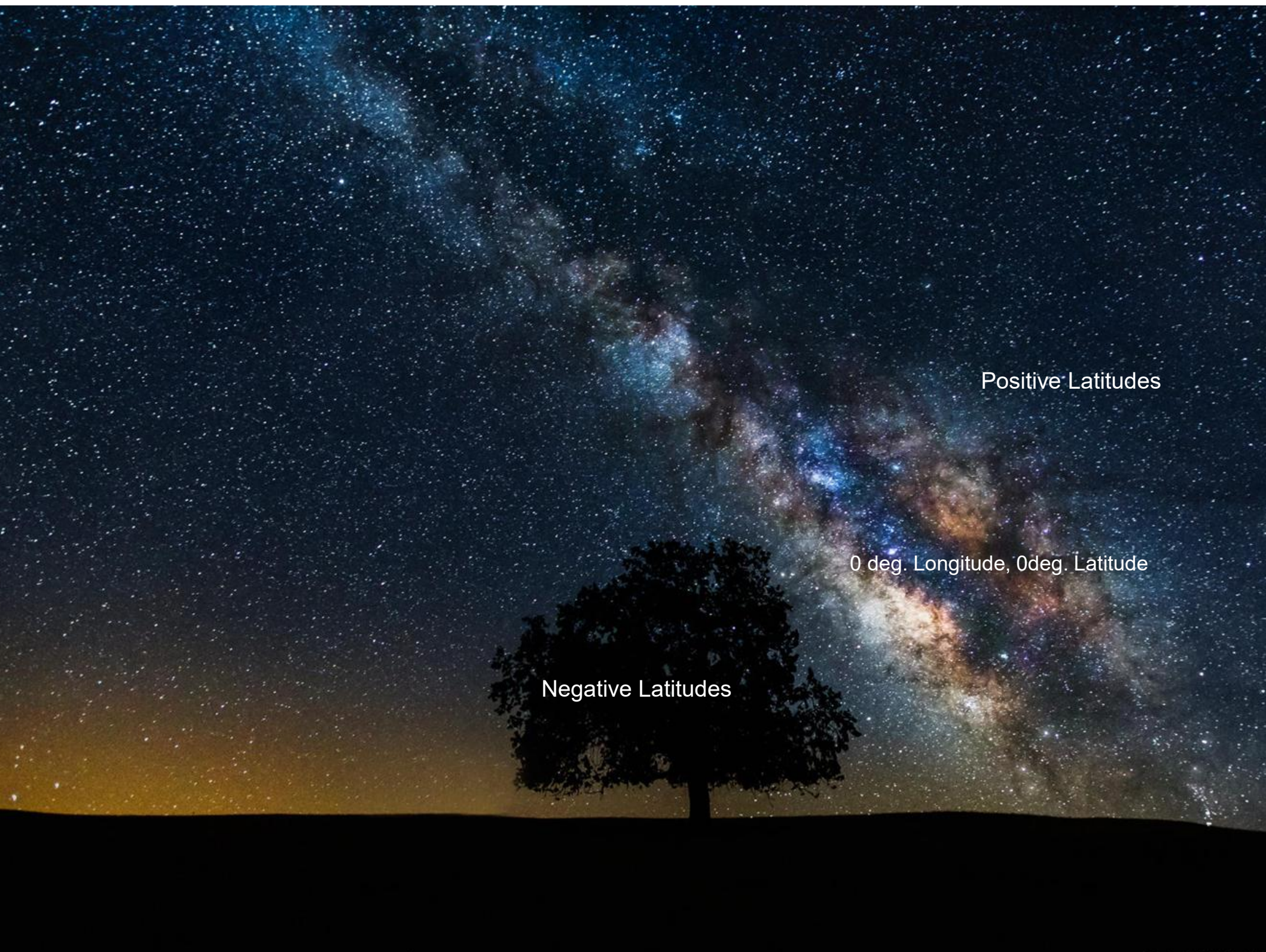
Galactic Coordinates Review









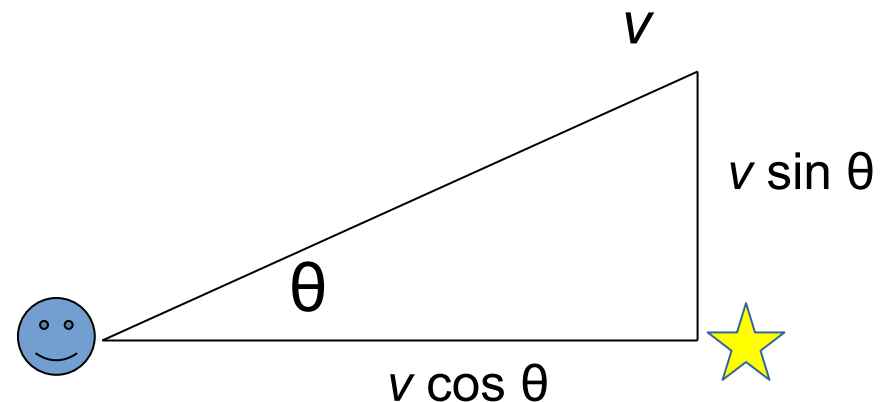


Relevant Motions

- As we learned in the first lecture, everything is in motion!
- The magnitude of the velocity shifts we measure depend on:
 - the direction of that motion with respect to the source's location
 - The maximum magnitude of the relative velocity
 - (so you can imagine there are lots of sines and cosines involved)

Relevant Motions

- We only care about **radial** motions!
- If θ is the angle between the motion of velocity v and the source, we are interested in
- $v_{\text{rad}} = v \cos \theta$
- the tangential component (which we cannot measure) is
- $v_{\text{tan}} = v \sin \theta$
[but of course we actually have two angles, the other one going into the screen....]



Relevant Motions

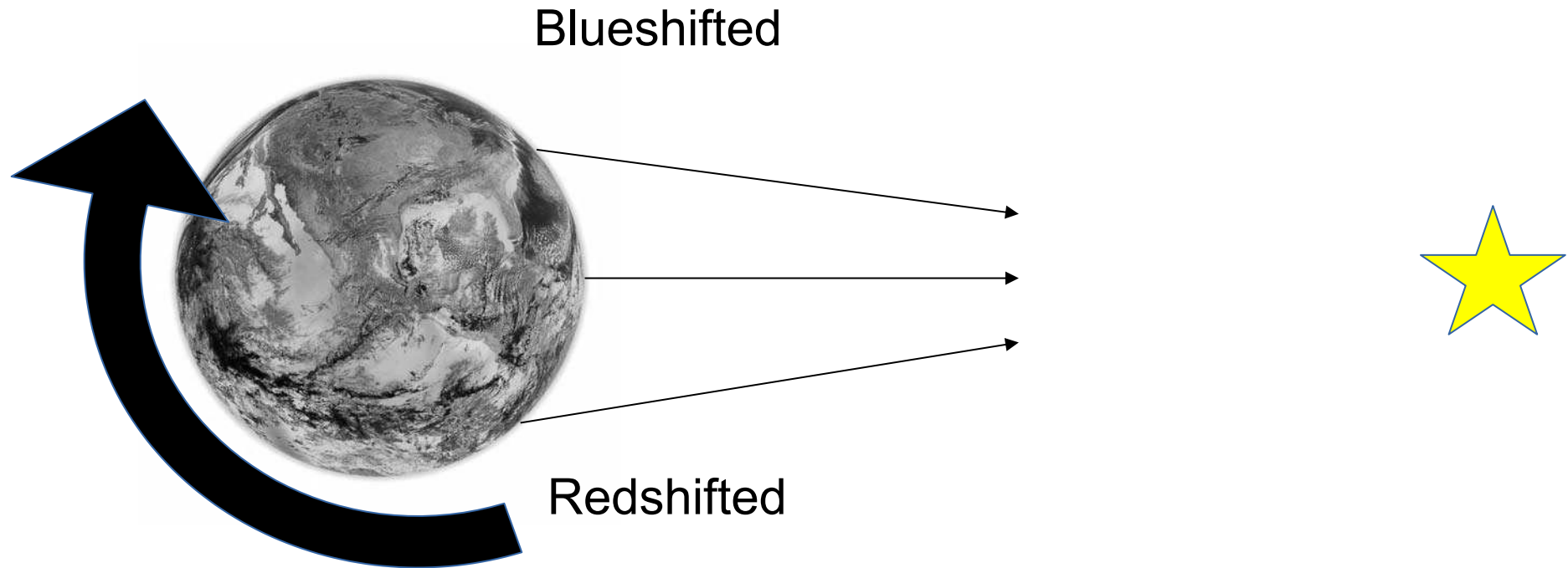
- 1) Earth rotation
 - $r = 6371\text{km}$
 - $<|0.5| \text{ km/s}$
- 2) Earth revolution around Sun
 - $r = 1\text{AU} = 1.5\text{e}8\text{km}$
 - $<|30| \text{ km/s}$
- 3) Sun non-circular motion
 - Found from observations of nearby stars/gas
 - $\sim|20| \text{ km/s}$
- 4) Sun (circular) motion around Galactic center **relative to circular motion of source**
 - generally $<|100| \text{ km/s}$

But First, Definitions: Blue and Red Shifts

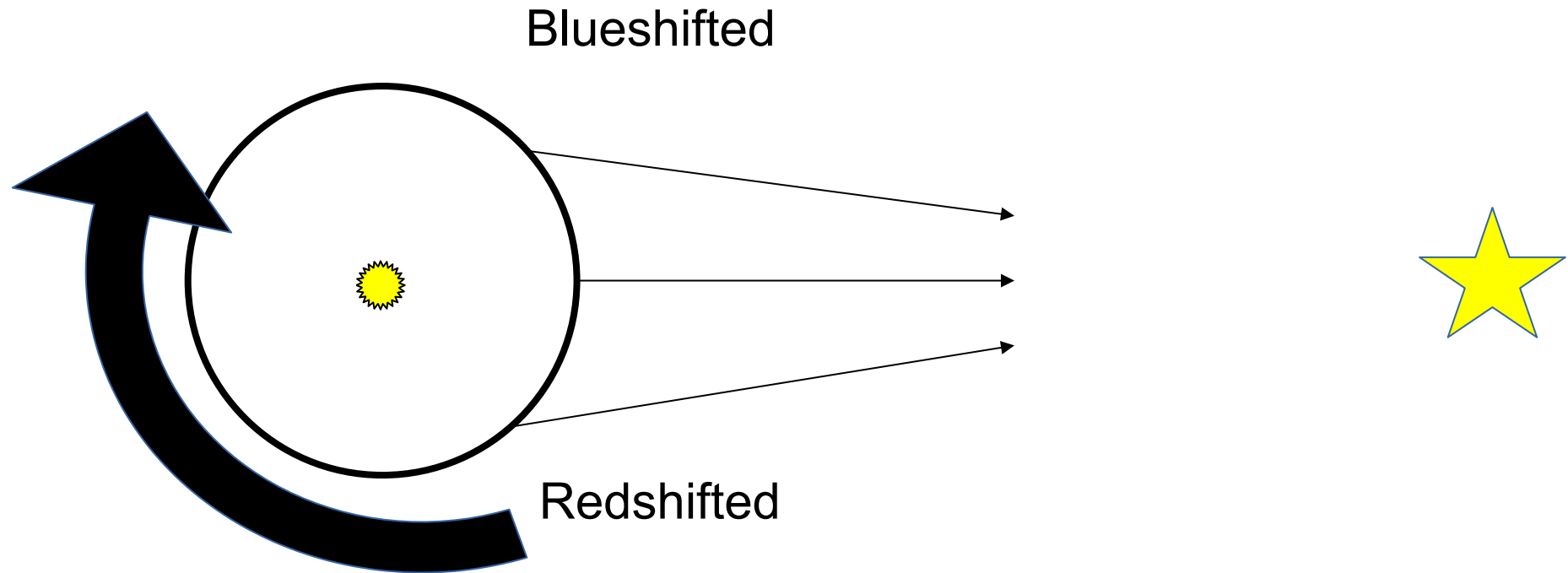
Blue-shifted means the object is moving toward the observer. We define this as having a negative radial velocity.

Redshifted means the object is moving away from the observer. We define this as having a positive radial velocity.

1. Earth Rotation



2. Earth Revolution



The Heliocentric Reference Frame

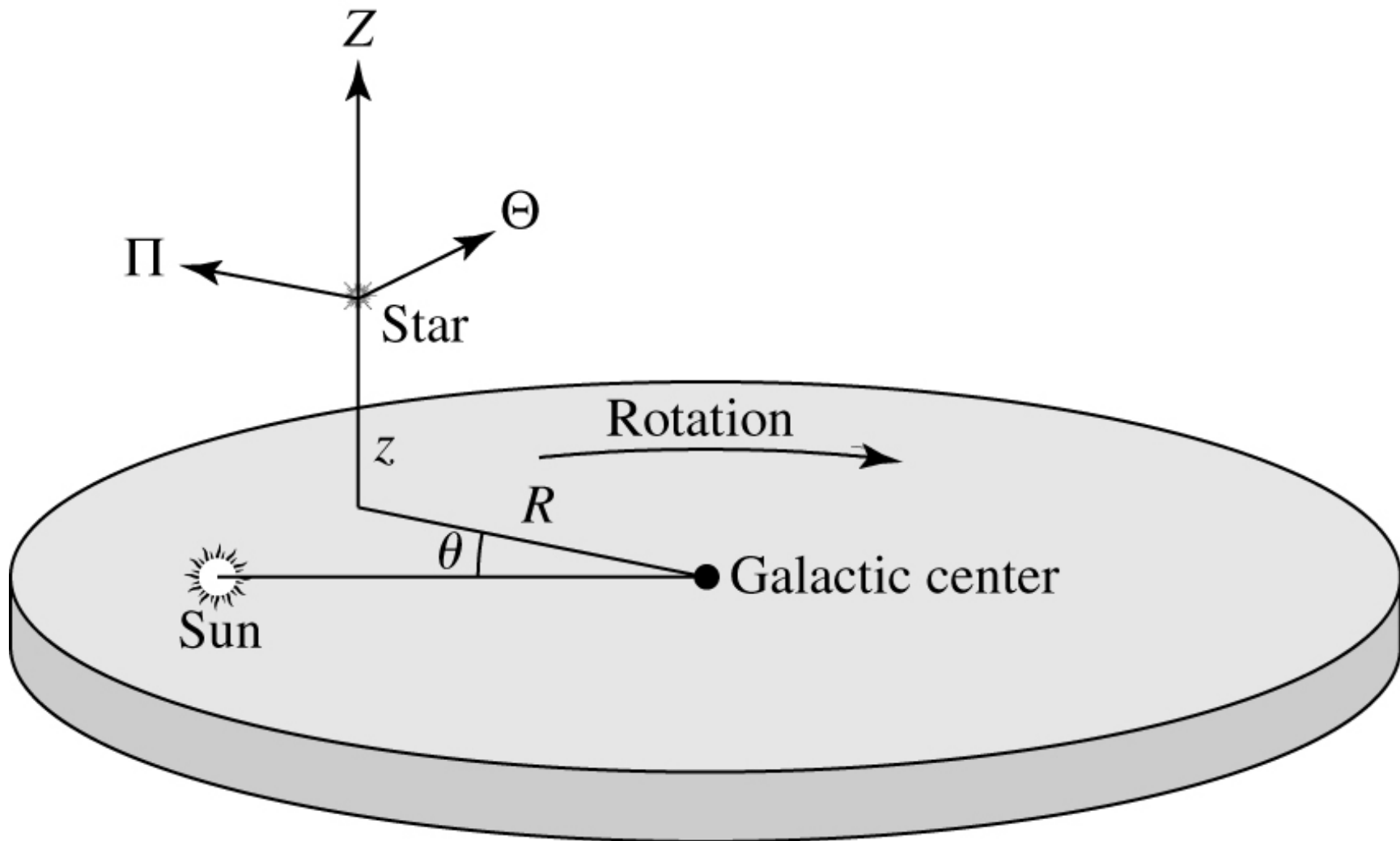
Correction for earth rotation and revolution puts us into the Heliocentric velocity frame.

Measurements in this velocity frame are what one would measure from the center of the Sun.

A related reference frame is “barycentric,” which instead uses the center of mass of the Solar System (essentially the same as Heliocentric).

These are frequently used for extragalactic observations, but we need to go one step further.

3. Sun Motion wrt Local Material



Local Standard of Rest (LSR)

We correct for these three effects so our measurements are in the local standard of rest (LSR) frame.

The LSR is the mean motion of the Sun about the Galactic center with respect to nearby material (stars and gas).

The goal of the LSR is to ensure that our measurements are comparable to those made at other times.

Local Standard of Rest (LSR)

The LSR frame therefore depends on:

- 1) The direction you are pointing your telescope
- 2) Your latitude on Earth
- 3) The time of day

These three combine to correct for the Earth's rotation

- 4) The time of year

This allows us to correct for the Earth's revolution

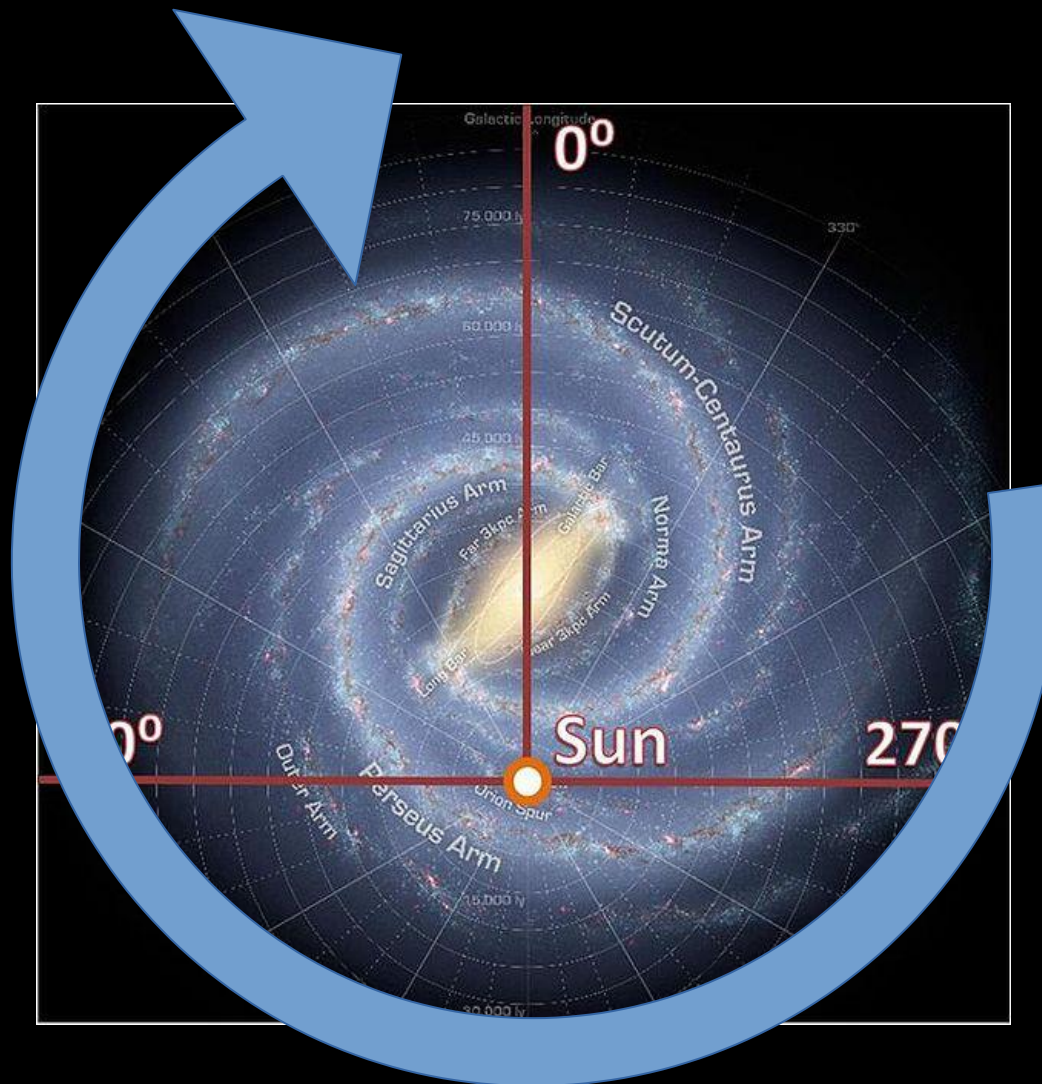
We can use the direction and the components of Solar rotation to correct to the LSR frame

<http://www.gb.nrao.edu/GBT/setups/radvelcalc.html>

4) LSR wrt Source Motion

THIS IS WHAT WE MEASURE IN THE LSR FRAME

Quadrant I

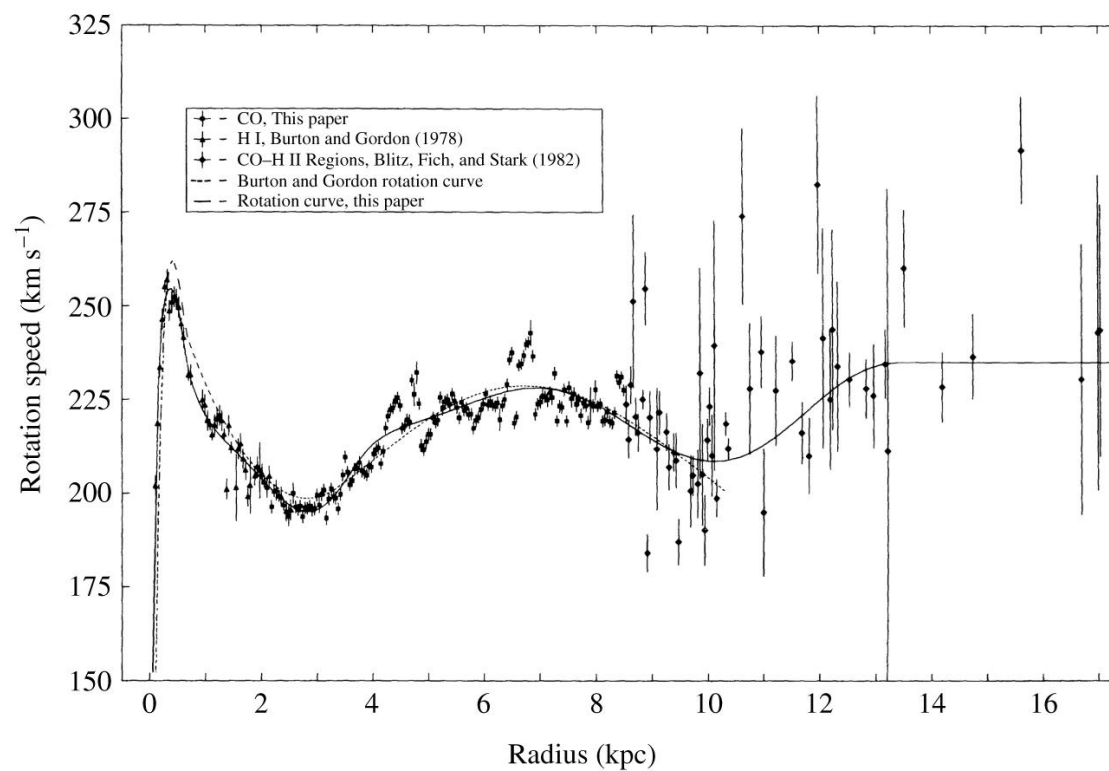


IV

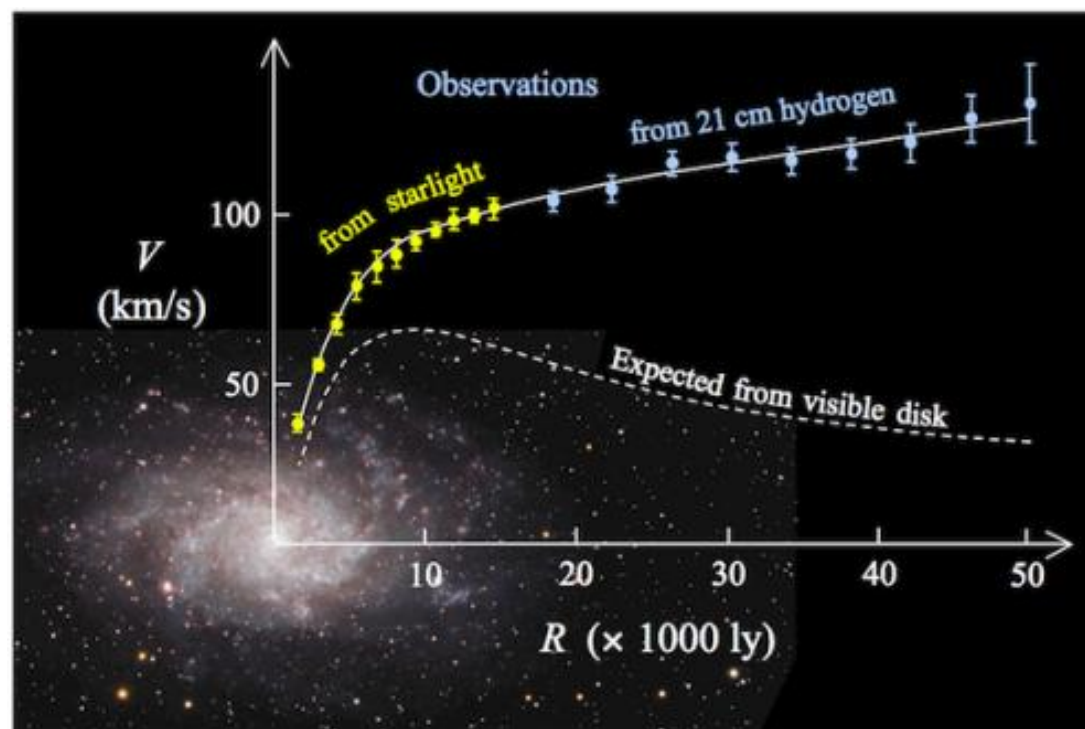
II

III

Milky Way



Other galaxy



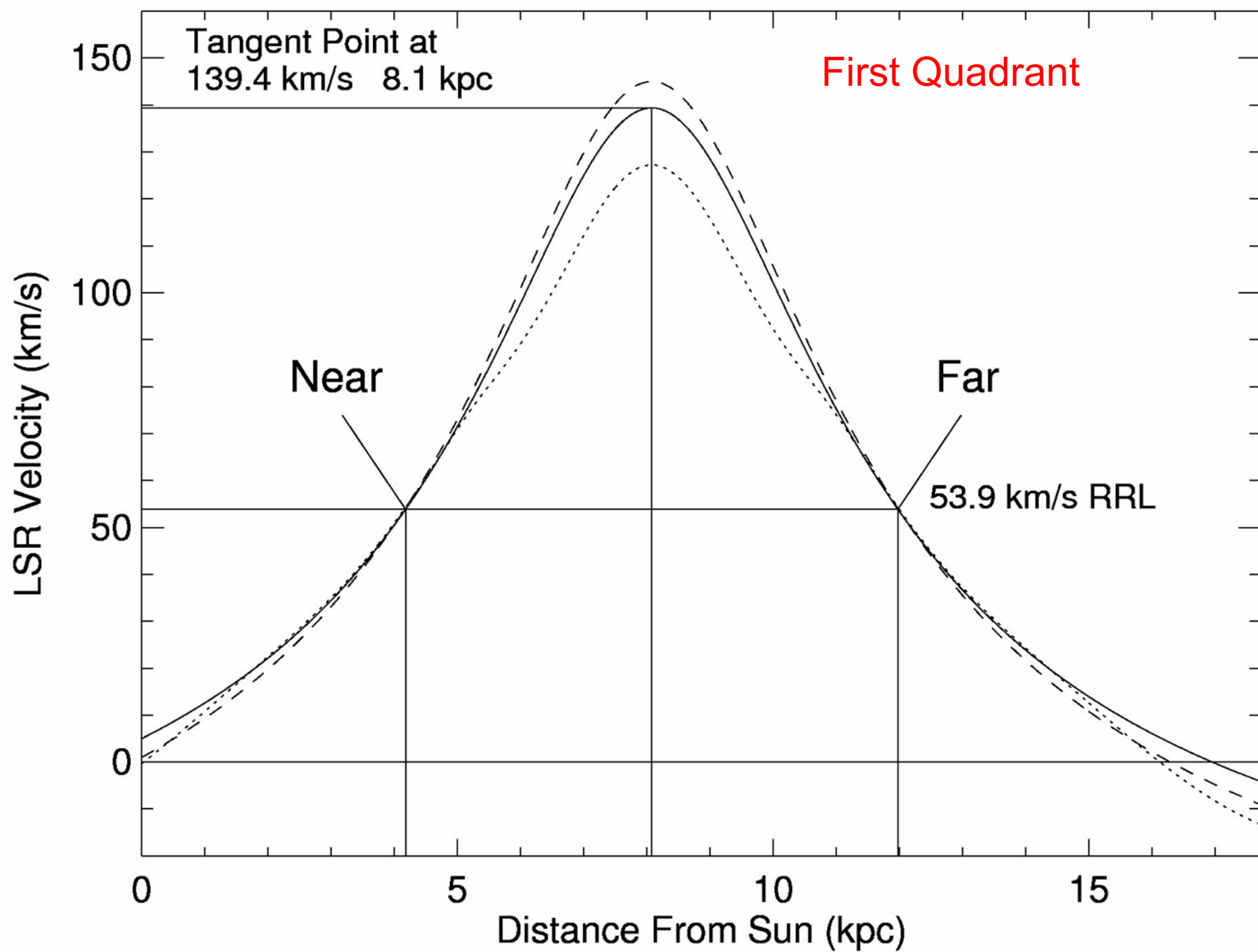
Radial Velocity Measurements

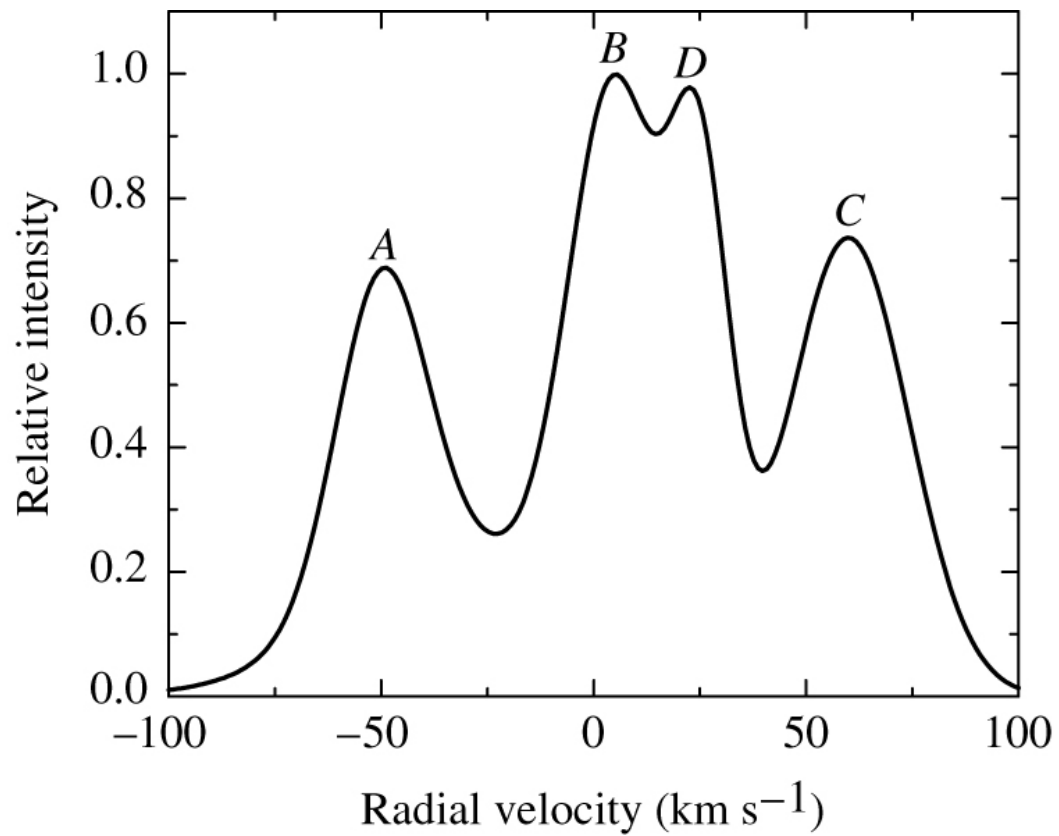
Although the rotation curve is flat, the radial velocity (in the LSR frame) depends on the **radial velocity** of the source wrt to the radial velocity of the Sun.

$$V_{\text{rad}} = (\omega - \omega_0) R_0 \sin l,$$

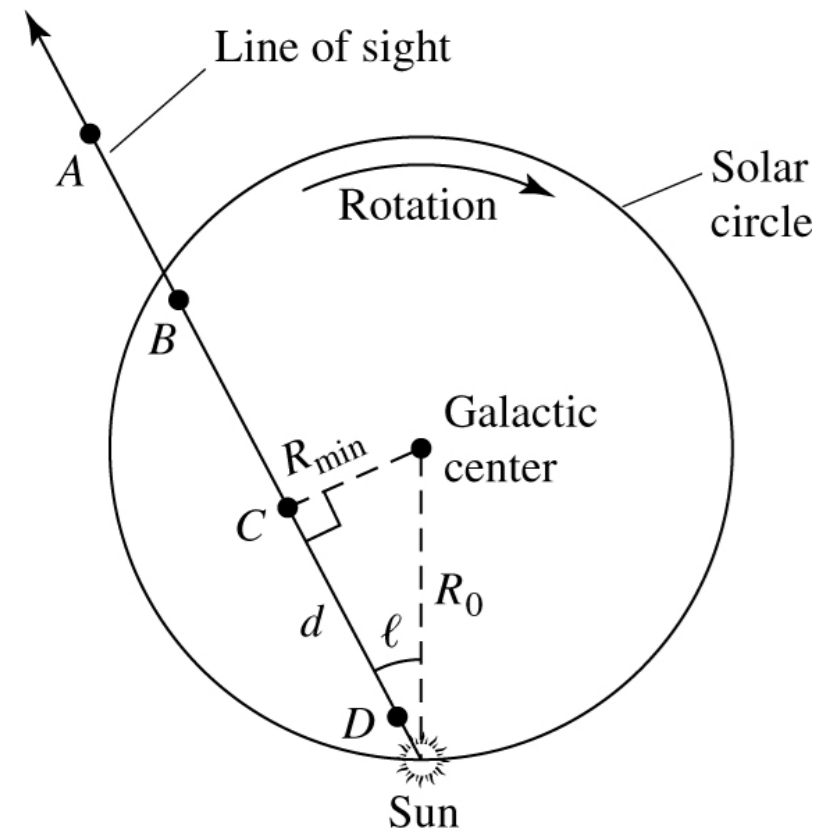
where ω is the angular velocity defined as v/R , R is the distance from the center of the Galaxy, l is the Galactic longitude, and the “0” subscripts denote Solar quantities. $R_0 = 8.3\text{kpc}$ and $\omega_0 = 29\text{ (km/s)/kpc}$.

-> We can use the measured radial velocity to determine R , the distance from the Galactic center to the source.





(a)



(b)

Longitude-Velocity Diagram

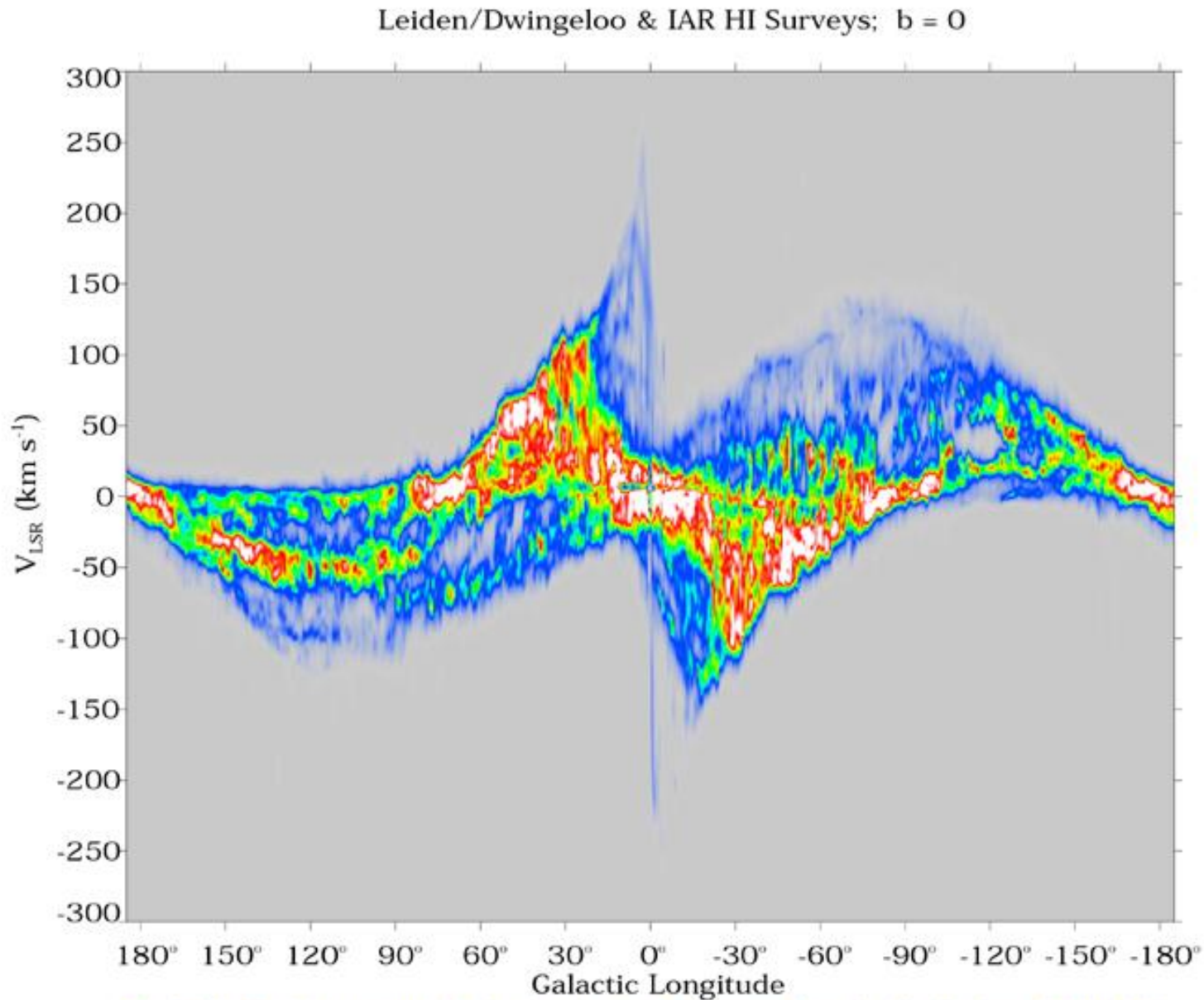
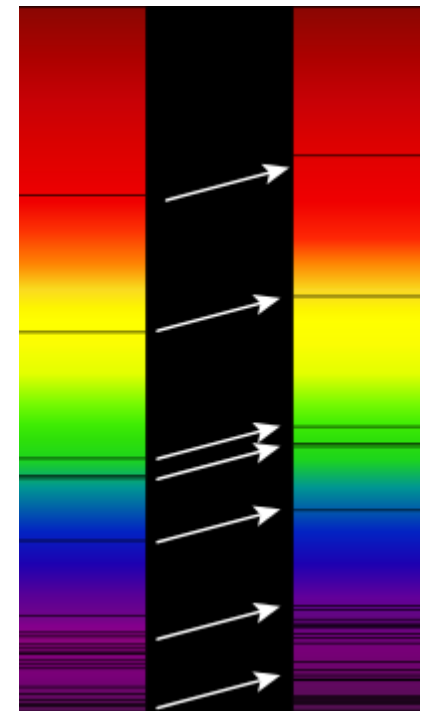
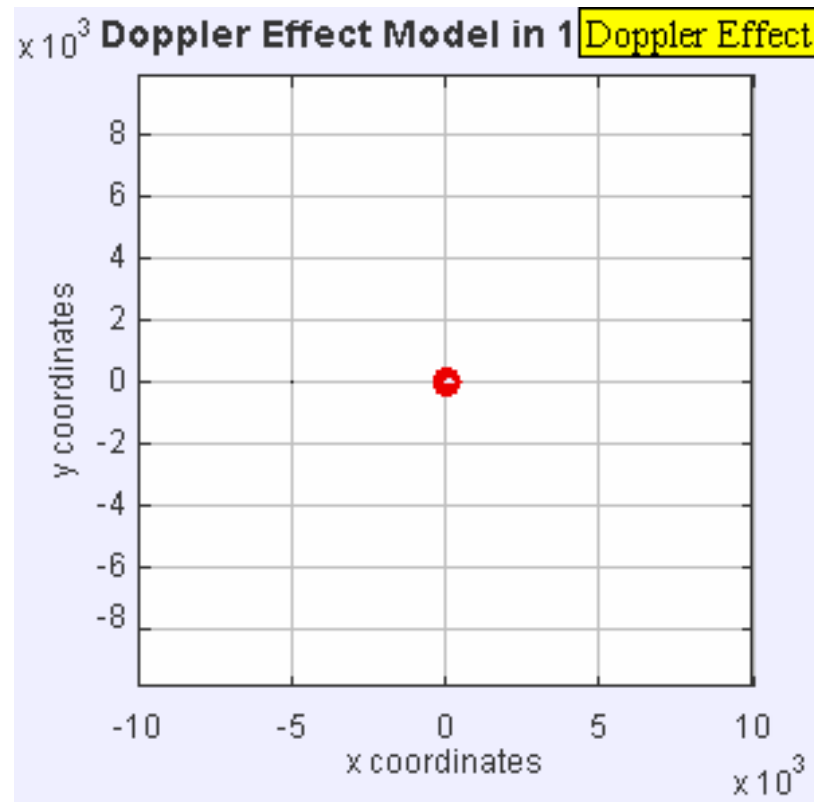


Fig 2.20 (D. Hartmann) 'Galaxies in the Universe' Sparke/Gallagher CUP 2007

Doppler shift (images from Wikipedia)



Doppler Shift

We **measure** emission at a particular frequency.

Due to the Doppler shift, we can convert frequency to velocity. Assume you are measuring a spectral line at $\nu_0 = 1420.4\text{MHz}$, but detect emission at $\nu = 1421.4\text{MHz}$. The Doppler shift equation is:

$$\nu = (\nu_0 - v)c/\nu_0 = \Delta\nu c/\nu_0$$

where v is the velocity, c is the speed of light ($c=3e5\text{km/s}$).

Therefore, $\Delta\nu = 1\text{MHz} = 1e6\text{Hz}$ and $v = 1e6\text{Hz} * 3e5 \text{ km/s} / 1420e6 \text{ Hz} = -211 \text{ km/s}$

This assumes no reference frame - it's just the relative radial velocity of the observatory wrt the source.